

TD01 - Linux and Shell Exercises

Exercises from CM-all-2025

Louis Ledoux

2026-07-05

Linux and Shell Exercises

Linux and Shell Exercises

Live pacing: about 8 min

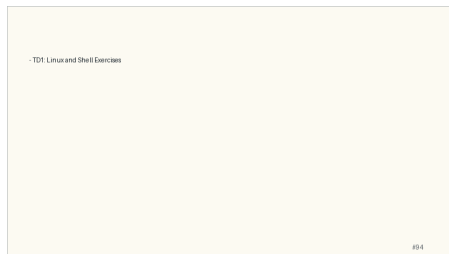
Question. What does the operating system do between the program and the machine?

Core claim. The OS gives a program controlled access to files, processes, memory, and devices.

Target capability. Students can place shell commands, processes, files, and permissions in the OS layer.

Main Path

- ▶ Start from the question: What does the operating system do between the program and the machine?
- ▶ Build the model: The OS gives a program controlled access to files, processes, memory, and devices.
- ▶ End with a checkable action: Students can place shell commands, processes, files, and permissions in the OS layer.
- ▶ Keep only this slide on the horizontal path during the live lecture.



Worked Example

- ▶ Use this as the live trace: Compare `ls`, an editor, and a compiled program: all are processes asking the OS for services.
- ▶ Representative source slide: 101
- ▶ Ask students to predict the next state before revealing the explanation.
- ▶ Then connect the result back to the core claim.

Ex8 Find

```
Describe the effect of the following instructions:  
find -name "gg*" -exec {} < /dev/null {} \;  
find /var/log -exec grep "www.athabasca.com" {} \;  
find -name "ccconf" -exec chmod o+r {} \;  
find /usr/local -type f -name "<*.html"
```

#101

Anecdote Or Motivation

A useful mental image: user programs are guests; the OS is the host that owns the keys.

- ▶ Use this only if students need a reason to care.
- ▶ If the room is already following, skip down to the check question.

Check Question

- ▶ Ask for a prediction, not a definition.
- ▶ Require a short justification: command trace, memory drawing, type rule, or file state.
- ▶ If more than a third of the room hesitates, go down to the source backup.

Ex8 Find

```
Describe the effect of the following instructions:  
find -name "qp2" -exec {} < {} in > {} out \;  
find /var/log -exec grep "www.athabasca.com" {} \;  
find -name "ccconf" -exec chmod o+r {} \;  
find /usr/local -type f -name "<" -html
```

#101

Backup Index

Original slides 94-101

- TD1: Linux and Shell Exercises

Backup: TD1: Linux and Shell Exercises

Original slide 94

- TD1: Linux and Shell Exercises

Backup: Ex1: Wildcard characters

Original slide 95

- ▶ List the entries inside the folder `/usr/bin` whose name starts with the letter `m`.
- ▶ List the entries inside the folder `/usr/bin` whose name starts with the letter `m` and it has exactly 3 characters
- ▶ List the entries inside the folder `/usr/bin` whose name starts with the letter `m` and it has at least 3 characters
- ▶ List the entries inside the folder `/usr/bin` whose name starts with the letter `m` and it has an extension (postfix after `.`)

Ex1 Wildcard characters

- List the entries inside the folder `/usr/bin` whose name starts with the letter `m`
- List the entries inside the folder `/usr/bin` whose name starts with the letter `m` and it has exactly 3 characters
- List the entries inside the folder `/usr/bin` whose name starts with the letter `m` and it has at least 3 characters
- List the entries inside the folder `/usr/bin` whose name starts with the letter `m` and it has an extension (postfix after `.`)

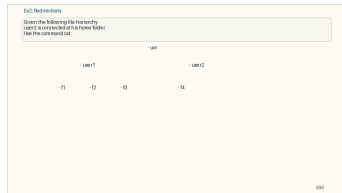
95

Backup: Ex2: Redirections

Original slide 96

Given the following file hierarchy
user2 is connected at his home folder
Use the command cat :
Create a file f5 with the following content
5 2 3 -1
Create a file f6 by concatenating f5 after f4
Add the file f1 of user1 at the file f6
Copy the file f2 of user1 as a new file f7 of user2

- ▶ usr
- ▶ user1
- ▶ user2
- ▶ f4
- ▶ f1
- ▶ f2
- ▶ f3



Backup: Ex3: Redirections

Original slide 97

- ▶ Create script with the name execution called with 3 arguments n1 n2 n3 (which are the filenames) performing the following manipulations
- ▶ P1 and P2 are programs that read from the stdin and print their result to stdout
- ▶ n1.r1
- ▶ P1
- ▶ n1
- ▶ n3
- ▶ P2
- ▶ n2.r1 concatenated with n1.r1
- ▶ n2.r1
- ▶ P1
- ▶ n2

Ex3 Redirections

- Create script with the name execution called with 3 arguments n1 n2 n3 (which are the filenames) performing the following manipulations
- P1 and P2 are programs that read from the stdin and print their result to stdout

-n1	-P1	-n1.r1			
-n2	-P1	-n2.r1	-n2.r1 concatenated with n1.r1	-P2	-n3

#97

Backup: Ex3: Redirections

Original slide 98

What happens when the following instruction is executed
`execution file1 ../user2/file2 fileN`

- ▶ usr
- ▶ user1
- ▶ user2
- ▶ file2
- ▶ file1
- ▶ p2
- ▶ file2
- ▶ p1
- ▶ Write an interactive version of the script execution that asks from the user to provide n1, n2 and n3, instead of passing them as arguments

Ex3: Redirections

What happens when the following instruction is executed in the home directory of user1?
`execution file1 ../user2/file2 fileN`

```
graph TD
    usr --> user1
    usr --> user2
    user1 --> file1
    user1 --> file2
    user2 --> p1
    user2 --> p2
    user2 --> file2
```

Write an interactive version of the script execution that asks from the user to provide n1, n2 and n3, instead of passing them as arguments.

4/88

Backup: Ex4: Script arguments

Original slide 99

- ▶ Create a script with the name argument using the command `expr` that prints one by one the arguments passed during the script call.
- ▶ For instance, the call:
- ▶ % argument one two three
- ▶ Should print in the stdout:
- ▶ 3 arguments :
- ▶ argument 1 - one
- ▶ argument 2 - two
- ▶ argument 3 - three

Ex4: Script arguments

- Create a script with the name argument using the command `expr` that prints one by one the arguments passed during the script call.
- For instance, the call:
- % argument one two three
- Should print in the stdout:
- 3 arguments:
- argument 1 - one
- argument 2 - two
- argument 3 - three

99

Backup: Ex5: Search

Original slide 100

- ▶ Create a script with the name mysearch passing two arguments during the call, i.e. motif and filename. The program prints either "filename has motif" or "filename does not have motif"
- ▶ Create a script mysearch2 filename that ask the user the motif to be searched inside the filename until the motif end is entered by the user
- ▶ Create a script mysearch3 f1 .. fn that performs the same thing as mysearch2, but now it searches inside the files given as arguments.
- ▶ /dev/null = trash

Ex5 Search

```
- Create a script with the name mysearch passing two arguments during the call, i.e. motif and filename. The program prints either "filename has motif" or "filename does not have motif".  
- Create a script mysearch2 filename that ask the user the motif to be searched inside the filename until the motif end is entered by the user.  
- Create a script mysearch3 f1 .. fn that performs the same thing as mysearch2, but now it searches inside the files given as arguments.  
-dev/null = trash
```

#100

Backup: Ex6: Find

Original slide 101

Describe the effect of the following instructions:

```
find ./ -name "tp2" -exec {} < f_in > {}.out \;  
find /var/log -exec grep "www.athabasca.com" {} \;  
find . -name "rc.conf" -exec chmod o+r {} \;  
find /usr/local -type f -name "*.html".
```

Ex6: Find

Describe the effect of the following instructions:
find -name "tp2" -exec {} < f_in > {}.out \;
find /var/log -exec grep "www.athabasca.com" {} \;
find . -name "rc.conf" -exec chmod o+r {} \;
find /usr/local -type f -name "*.html".

#101